Problem Statement – Data-Driven Pomegranate Procurement Strategy

Nov 5, 2026 · @deepti

1. Business Context

Buying farm produce is never simple. Mandi prices keep changing, supply is uncertain, and seasons, weather and transport costs all add to the challenge. Pomegranates in Maharashtra, India's top-producing state, show this well: the state's agriculture minister names disease, unpredictable weather and unstable prices as grower problems [1], heavy rain hurt the 2025-26 crop [2], and buyers move to Gujarat and Rajasthan when Maharashtra's April–December season ends [3]. This project uses pomegranate buying in Maharashtra as its business example.

2. Problem Statement 1 – Limited Visibility into Future Price and Market Availability

Problem	Buying teams cannot see clearly what prices and supply will look like in each district, so they have to decide without a reliable view of what comes next.
Evidence	**Project analysis:** monthly average pomegranate prices ranged from about ₹4,500 to ₹12,250 per quintal, and the amount of fruit arriving in markets differed a lot by district and season. **External source:** weather disrupted supply in the 2025-26 season [2].
Business question	"How can historical mandi prices, arrivals and weather information be used to estimate next-period pomegranate prices and market availability across sourcing districts?"
Stakeholders	Procurement managers, sourcing teams, suppliers/farmers and downstream buyers

3. Problem Statement 2 – Difficulty in Selecting the Optimal Sourcing Mix

Problem	Even with a good price and supply forecast, the team still has to decide where to buy and how much. That choice must balance price, available supply, transport cost and the risk of relying too heavily on one district.
Evidence	**Illustrative scenario, not an actual company order:** the project uses a 1,000-tonne pomegranate requirement to test how the quantity could be split across districts. Districts differ in price, supply and distance, so the cheapest mandi price does not always mean the lowest delivered cost, and buying from just one district adds risk.
Business question	"How can predicted price, expected availability and logistics cost be converted into an optimal district-wise procurement allocation while managing sourcing constraints?"
Stakeholders	Procurement managers, supply-chain/logistics teams, finance teams, suppliers and business leadership

4. Project Outcome

Price & Availability Prediction → Expected Landed Cost → Procurement Optimization → District-wise Recommendation → Decision-Support Dashboard

Sources

Figures marked "Project analysis" come from the project's own mandi price and arrivals data. All other facts come from the sources below.

1. Free Press Journal – Maharashtra aims to make Indian pomegranates a global brand (Sep 2026)
2. FreshPlaza – Indian pomegranate supply stays short while growers prepare for stricter export markets (2026)
3. FreshPlaza – India's pomegranate supply shifts as Maharashtra's season ends